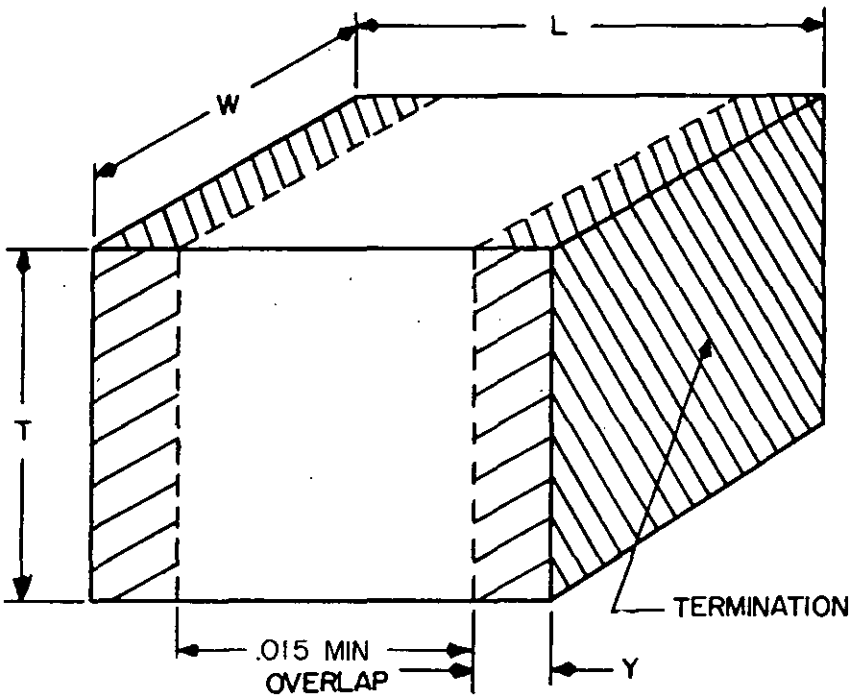
MILITARY SPECIFICATION SHEET

CAPACITORS, CHIP, MULTIPLE LAYER, FIXED, UNENCAPSULATED, CERAMIC DIELECTRIC, ESTABLISHED RELIABILITY, STYLES CDR11, CDR12, CDR13, AND CDR14 (HIGH FREQUENCY)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the capacitors described herein shall consist of this document and the latest issue of Specification MIL-C-55681.



INCHES	MM
.005	.13
.010	.25
.015	.38
.020	.51
.028	.71
.055	1.40
.057	1.45
.058	1.47
.102	2.59
.110	2.79

Dimensions					
Style	L	W	T		Y
			MIN	MAX	
CDR11	.055 ±.015	.055 ±.015	.028	.057	.010 ±.005
CDR12	.055 ±.025	.055 ±.015	.028	.057	.010 ±.005
CDR13	.110 ±.020	.110 ±.020	.058	.102	.015 ±.010
CDR14	.110 ±.025	.110 ±.020	.058	.102	.015 ±.010

APPROXIMATE ACTUAL SIZE FOR STYLES CDR11 AND CDR12

APPROXIMATE ACTUAL SIZE FOR STYLES CDR13 AND CDR14

NOTES:
1. Dimensions are in inches.
2. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.

FSC 5910

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TABLE I. Styles CDR11 and CDR12 capacitor characteristics.

Type designation 1/	Capacitance	Capacitance tolerance	Rated temperature and voltage-temperature limits	Rated voltage
	pF			volts, dc
CDR1-B-0R1AB--	0.1	B	BG, BP	50
CDR1-B-0R2AB--	0.2	B		
CDR1-B-0R3A---	0.3	B,C		
CDR1-B-0R4A---	0.4	B,C		
CDR1-B-0R5A---	0.5	B,C,D		
CDR1-B-0R6A---	0.6			
CDR1-B-0R7A---	0.7			
CDR1-B-0R8A---	0.8			
CDR1-B-0R9A---	0.9			
CDR1-B-1R0A---	1.0			
CDR1-B-1R1A---	1.1			
CDR1-B-1R2A---	1.2			
CDR1-B-1R3A---	1.3			
CDR1-B-1R4A---	1.4			
CDR1-B-1R5A---	1.5			
CDR1-B-1R6A---	1.6			
CDR1-B-1R7A---	1.7			
CDR1-B-1R8A---	1.8			
CDR1-B-1R9A---	1.9			
CDR1-B-2R0A---	2.0			
CDR1-B-2R1A---	2.1			
CDR1-B-2R2A---	2.2			
CDR1-B-2R4A---	2.4			
CDR1-B-2R7A---	2.7			
CDR1-B-3R0A---	3.0			
CDR1-B-3R3A---	3.3			
CDR1-B-3R6A---	3.6			
CDR1-B-3R9A---	3.9			
CDR1-B-4R3A---	4.3			
CDR1-B-4R7A---	4.7			
CDR1-B-5R1A---	5.1			
CDR1-B-5R6A---	5.6			
CDR1-B-6R2A---	6.2			
CDR1-B-6R8A---	6.8	B,C,J,K,M		
CDR1-B-7R5A---	7.5			
CDR1-B-8R2A---	8.2			
CDR1-B-9R1A---	9.1			
CDR1-B-100A---	10	F,G,J,K,M		
CDR1-B-110A---	11			
CDR1-B-120A---	12			
CDR1-B-130A---	13			
CDR1-B-150A---	15			
CDR1-B-160A---	16			
CDR1-B-180A---	18			
CDR1-B-200A---	20			
CDR1-B-220A---	22			
CDR1-B-240A---	24			
CDR1-B-270A---	27			

1/ See footnote at end of table.

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TABLE I. Styles CDR11 and CDR12 capacitor characteristics. -Continued.

Type designation 1/	Capacitance	Capacitance tolerance	Rated temperature and voltage-temperature limits	Rated votlage
	pF			volts,dc
CDR1-B-300A---	30	F,G,J,K,M	BG, BP	50
CDR1-B-330A---	33			
CDR1-B-360A---	36			
CDR1-B-390A---	39			
CDR1-B-430A---	43			
CDR1-B-470A---	47			
CDR1-B-510A---	51			
CDR1-B-560A---	56			
CDR1-B-620A---	62			
CDR1-B-680A---	68			
CDR1-B-750A---	75			
CDR1-B-820A---	82			
CDR1-B-910A---	91			
CDR1-B-101A---	100			
CDR1-BP111E---	110		BP	
CDR1-BP121E---	120			
CDR1-BP131E---	130			
CDR1-BP151E---	150			
CDR1-BP161E---	160			
CDR1-BP181E---	180			
CDR1-BP201E---	200			
CDR1-BP221E---	220			
CDR1-BP241E---	240			
CDR1-BP271E---	270			
CDR1-BP301E---	300			
CDR1-BP331E---	330			
CDR1-BP361E---	360			
CDR1-BP391E---	390			
CDR1-BP431E---	430			
CDR1-BP471E---	470			
CDR1-BP511E---	510			
CDR1-BP561E---	560			
CDR1-BP621E---	620			
CDR1-BP681E---	680			
CDR1-BP751E---	750			
CDR1-BP821E---	820			
CDR1-BP911E---	910			
CDR1-BP102E---	1000			

1/ Complete type designation will include additional symbols to indicate style, voltage-temperature limits, capacitance tolerance (where applicable), termination finish ("M" for style CDR11, and "S" for style CDR12) and failure rate level.

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TABLE II. Styles CDR13 and CDR14 capacitor characteristics.

Type designation 1/	Capacitance	Capacitance tolerance	Rated temperature and voltage--- temperature limits	Rated voltage
	pF			volts, dc
CDR1-B-0R1EB---	0.1	B	BG, BP	500
CDR1-B-0R2EB---	0.2	B		
CDR1-B-0R3E---	0.3	B,C		
CDR1-B-0R4E---	0.4	B,C		
CDR1-B-0R5E---	0.5	B,C,D		
CDR1-B-0R6E---	0.6			
CDR1-B-0R7E---	0.7			
CDR1-B-0R8E---	0.8			
CDR1-B-0R9E---	0.9			
CDR1-B-1R0E---	1.0			
CDR1-B-1R1E---	1.1			
CDR1-B-1R2E---	1.2			
CDR1-B-1R3E---	1.3			
CDR1-B-1R4E---	1.4			
CDR1-B-1R5E---	1.5			
CDR1-B-1R6E---	1.6			
CDR1-B-1R7E---	1.7			
CDR1-B-1R8E---	1.8			
CDR1-B-1R9E---	1.9			
CDR1-B-2R0E---	2.0			
CDR1-B-2R1E---	2.1			
CDR1-B-2R2E---	2.2			
CDR1-B-2R4E---	2.4			
CDR1-B-2R7E---	2.7			
CDR1-B-3R0E---	3.0			
CDR1-B-3R3E---	3.3			
CDR1-B-3R6E---	3.6			
CDR1-B-3R9E---	3.9			
CDR1-B-4R3E---	4.3			
CDR1-B-4R7E---	4.7			
CDR1-B-5R1E---	5.1			
CDR1-B-5R6E---	5.6			
CDR1-B-6R2E---	6.2			
CDR1-B-6R8E---	6.8	B,C,J,K,M		
CDR1-B-7R5E---	7.5			
CDR1-B-8R2E---	8.2			
CDR1-B-9R1E---	9.1			
CDR1-B-100E---	10	F,G,J,K,M		
CDR1-B-110E---	11			
CDR1-B-120E---	12			
CDR1-B-130E---	13			
CDR1-B-150E---	15			
CDR1-B-160E---	16			
CDR1-B-180E---	18			
CDR1-B-200E---	20			
CDR1-B-220E---	22			
CDR1-B-240E---	24			
CDR1-B-270E---	27			
CDR1-B-300E---	30			
CDR1-B-330E---	33			
CDR1-B-360E---	36			
CDR1-B-390E---	39			
CDR1-B-430E---	43			
CDR1-B-470E---	47			
CDR1-B-510E---	51			

1/ See footnote at end of table.

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TABLE II. Styles CDR13 and CDR14 capacitor characteristics. -Continued.

Type designation 1/	Capacitance	Capacitance tolerance	Rated temperature and voltage-temperature limits	Rated voltage
	pF			volts, dc
CDR1-B-560E---	56	F,G,J,K,M,	BG, BP	500
CDR1-B-620E---	62			
CDR1-B-680E---	68			
CDR1-B-750E---	75			
CDR1-B-820E---	82			
CDR1-B-910E---	91			
CDR1-B-101E---	100			
CDR1-B-111D---	110			300
CDR1-B-121D---	120			
CDR1-B-131D---	130			
CDR1-B-151D---	150			
CDR1-B-161D---	160			
CDR1-B-181D---	180			
CDR1-B-201D---	200			
CDR1-B-221C---	220			200
CDR1-B-241C---	240			
CDR1-B-271C---	270			
CDR1-B-301C---	300			
CDR1-B-331C---	330			
CDR1-B-361C---	360			
CDR1-B-391C---	390			
CDR1-B-431C---	430			
CDR1-B-471C---	470			
CDR1-B-511B---	510			100
CDR1-B-561B---	560			
CDR1-B-621B---	620			
CDR1-B-681A---	680			50
CDR1-B-751A---	750			
CDR1-B-821A---	820			
CDR1-B-911A---	910			
CDR1-B-102A---	1000			
CDR1-BP112A---	1100		BP	
CDR1-BP122A---	1200			
CDR1-BP132A---	1300			
CDR1-BP152A---	1500			
CDR1-BP162A---	1600			
CDR1-BP182A---	1800			
CDR1-BP202A---	2000			
CDR1-BP222A---	2200			
CDR1-BP242A---	2400			
CDR1-BP272A---	2700			
CDR1-BP302A---	3000			
CDR1-BP332A---	3300			
CDR1-BP362A---	3600			
CDR1-BP392A---	3900			
CDR1-BP432A---	4300			
CDR1-BP472A---	4700			
CDR1-BP502A---	5000			
CDR1-BP512A---	5100			

1/ Complete type disignation will include additional symbols to indicate style, voltage-temperature limits, capacitance tolerance (where applicable), termination finish, ("M" for style CDR 13 and "S" for style CDR14) and failure rate level.

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REQUIREMENTS:

Dimensions and configuration: See figure.

Case type: Multi-layer, unencapsulated, monolithic.
Material: Porcelain or ceramic.

Terminations:
Styles CDR11 and CDR13: Metallized.
Styles CDR12 and CDR14: Solder coated, 372°F, 62 Sn, 36Pb 2 percent Ag.

Inspection conditions: All visual examinations shall be performed using a stereo microscope having a minimum magnification of 20 power.

Capacitance: See tables I and II.

Capacitance tolerance: See tables I and II.

Rated voltage: See tables I and II.

Operating temperature range: -55°C to +125°C.

Insulation resistance:
At +25° ±5°C: 10^6 megohms, minimum (for 0.1 pF thru 470 pF).
10^5 megohms, minimum (for 510 pF thru 1,000 pF).

At +125° ±5°C: 10^5 megohms, minimum (for 0.1 pF thru 470 pF).
10^4 megohms, minimum (for 510 pF thru 1,000 pF).

Quality factor: In accordance with MIL-C-55681.

Resonance: In accordance with MIL-C-55681.

NOTE: These capacitors are intended for use in RF solid-state circuitry up to and including 25 gigahertz, and cannot be replaced in the field.

NOTE: Revision letters are not used to denote changes due to the extensiveness of the changes.

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Navy - EC
Air Force - 11

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Navy - AS, OS
Air Force - 11, 17, 85
DSA - ES

User activities:
Army -
Navy - MC, SH
Air Force - 19

Preparing activity:
Army - ER

Agent:
DSA - ES

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